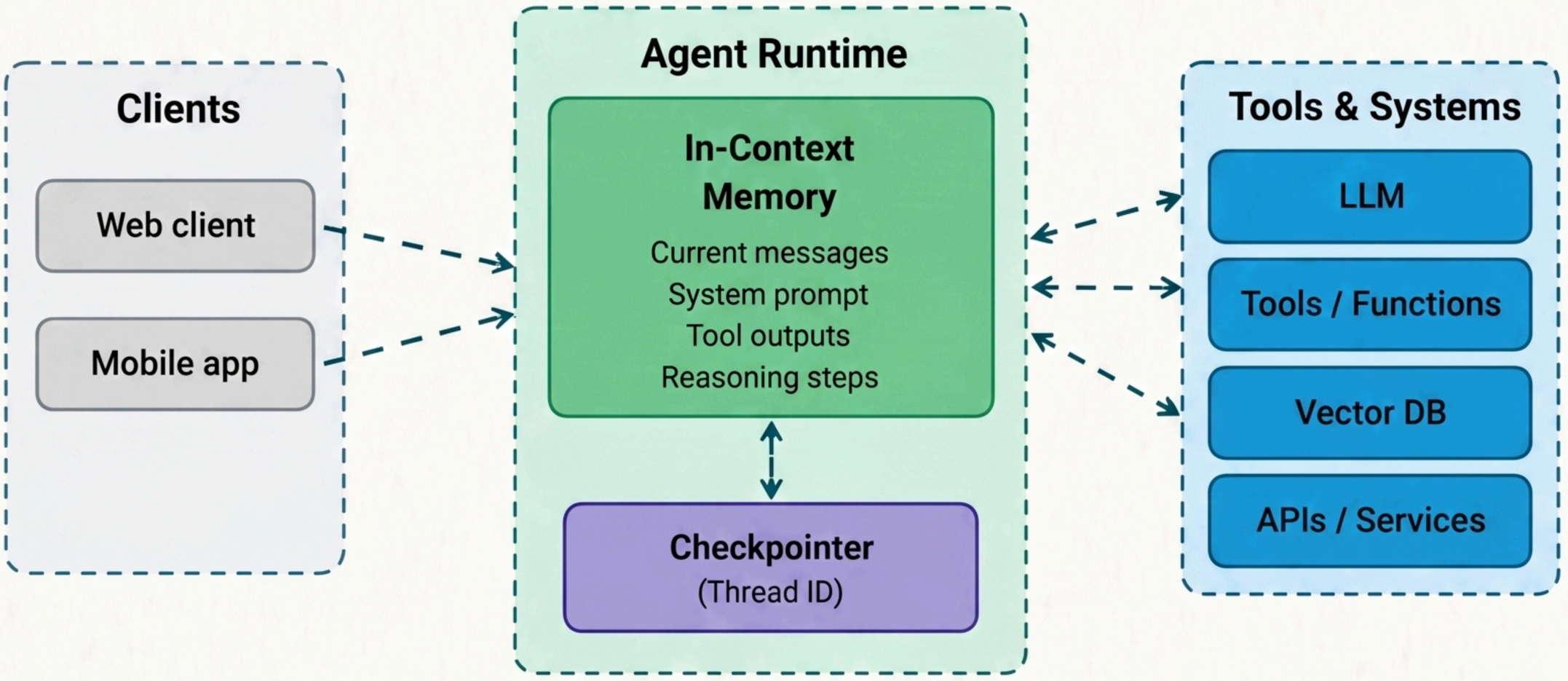
In-Context / Working Memory

(In-Context Scope)



WHAT IT IS The active context window — everything the model can “see” right now.

WHY IT'S USED Agent’s scratchpad to keep conversations and steps coherent.

WHERE IT’S USED All multi-turn agents / chatbots, managed via a checkpointer.

EXAMPLE Ask “What did I just say?” — it looks back in the context and replies.

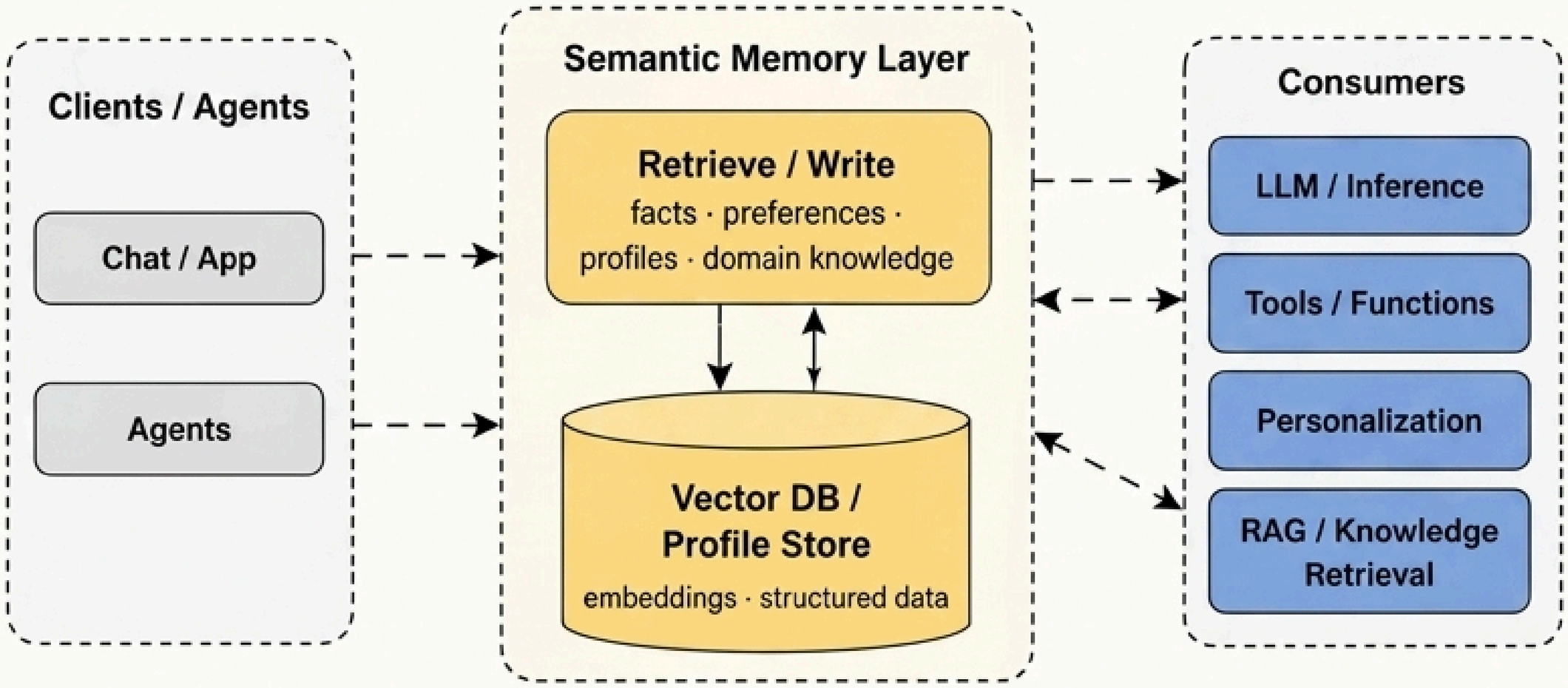
▶ **INTERVIEW SIGNAL** @DEVENDHIRAN

“How do you manage context across turns?”

→ Look for context window management, checkpointing, and truncation strategy.

Semantic Memory

Knowledge that lasts.



WHAT IT IS Persistent store of facts, preferences, and domain knowledge about users or topics.

WHY IT'S USED Enables personalized, consistent, and accurate responses across sessions.

WHERE IT'S USED Personalized assistants, support bots, and enterprise knowledge agents.

EXAMPLE Agent remembers "user prefers metric units" and applies it next time.

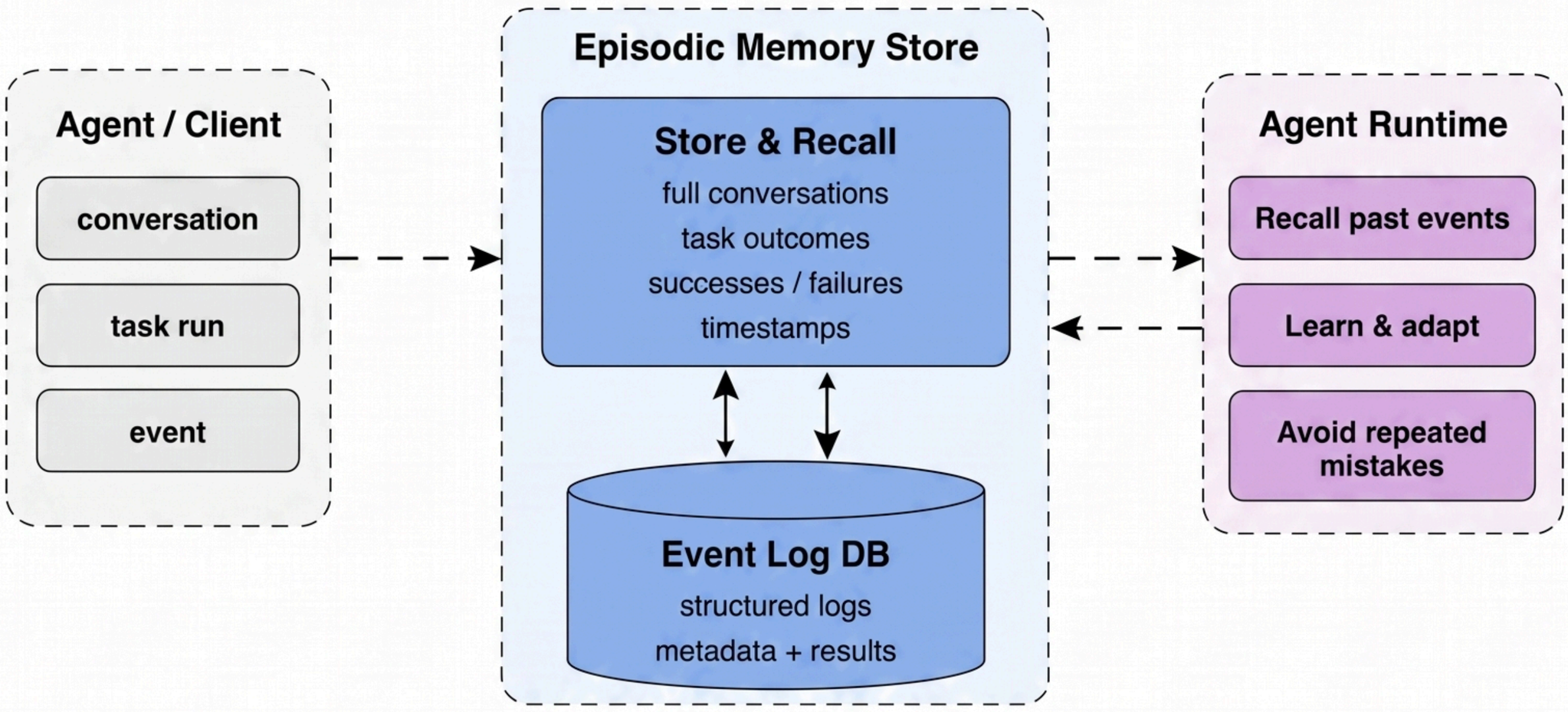
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"How do you design long-term memory for an AI agent?"

→ Look for data model, storage choice, retrieval strategy, and privacy considerations.

Episodic Memory

Experiences that teach.



WHAT IT IS A record of past events, conversations, and task outcomes.

WHY IT'S USED Agents learn from experience and avoid repeating failures.

WHERE IT'S USED Coding agents, decision agents, and support systems needing audit trails.

EXAMPLE Reflexion (Shinn et al., NeurIPS 2023): agent writes a self-reflection after each failed attempt, stores it, and uses it next time — achieving 91% accuracy vs GPT-4's 80% on a coding benchmark.

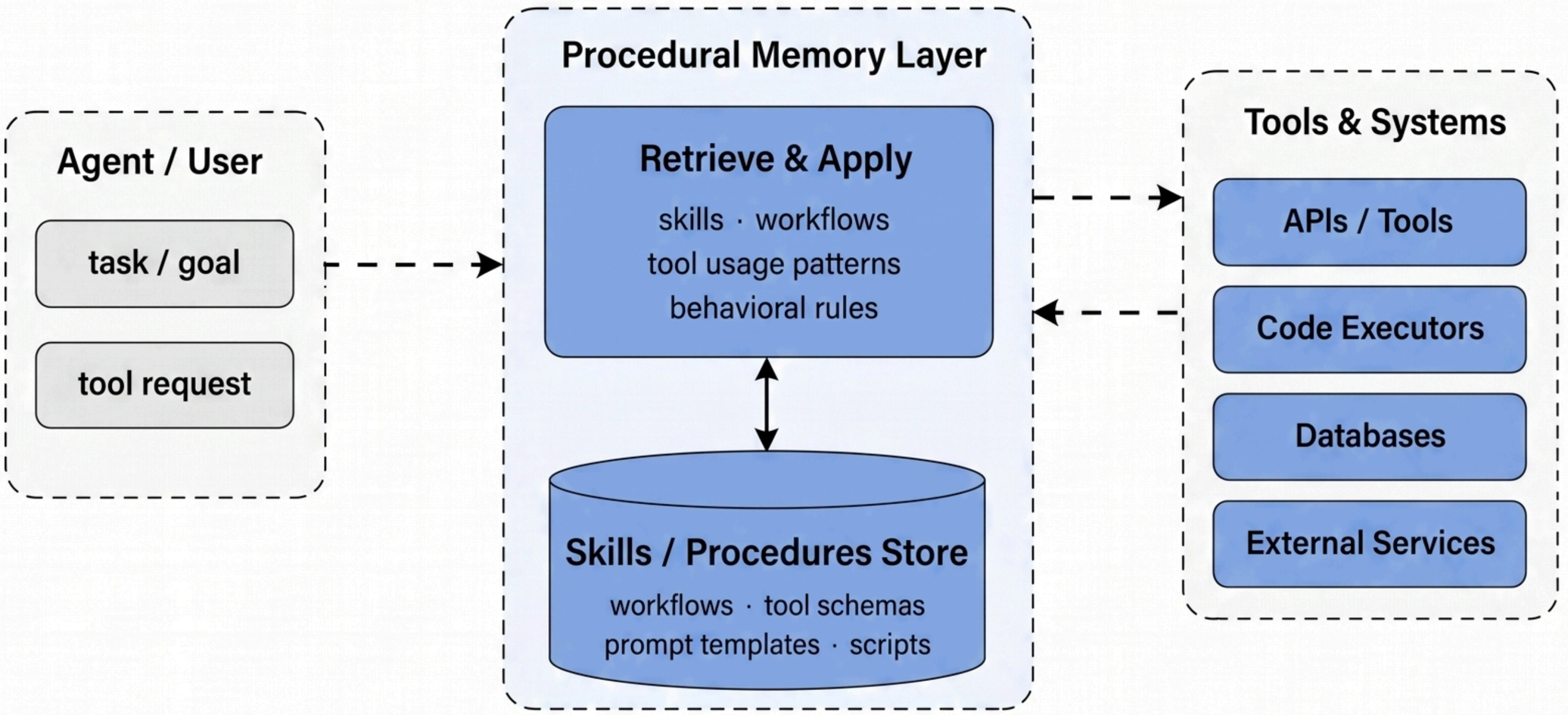
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"How would you design memory so agents can learn from past failures?"

→ Look for event logging, retrieval by similarity, summarization, and retention policies.

Procedural Memory

Knowledge of how to do things.



WHAT IT IS Knowledge of how tasks are done — skills, workflows, and rules.

WHY IT'S USED Agents follow known procedures instead of re-deriving steps every time.

WHERE IT'S USED Workflow automation, coding agents; stored in prompts, tool registries, and skill libs.

EXAMPLE Voyager (Minecraft agent) builds a library of executable skills.
For new tasks, it reuses and composes existing skills.

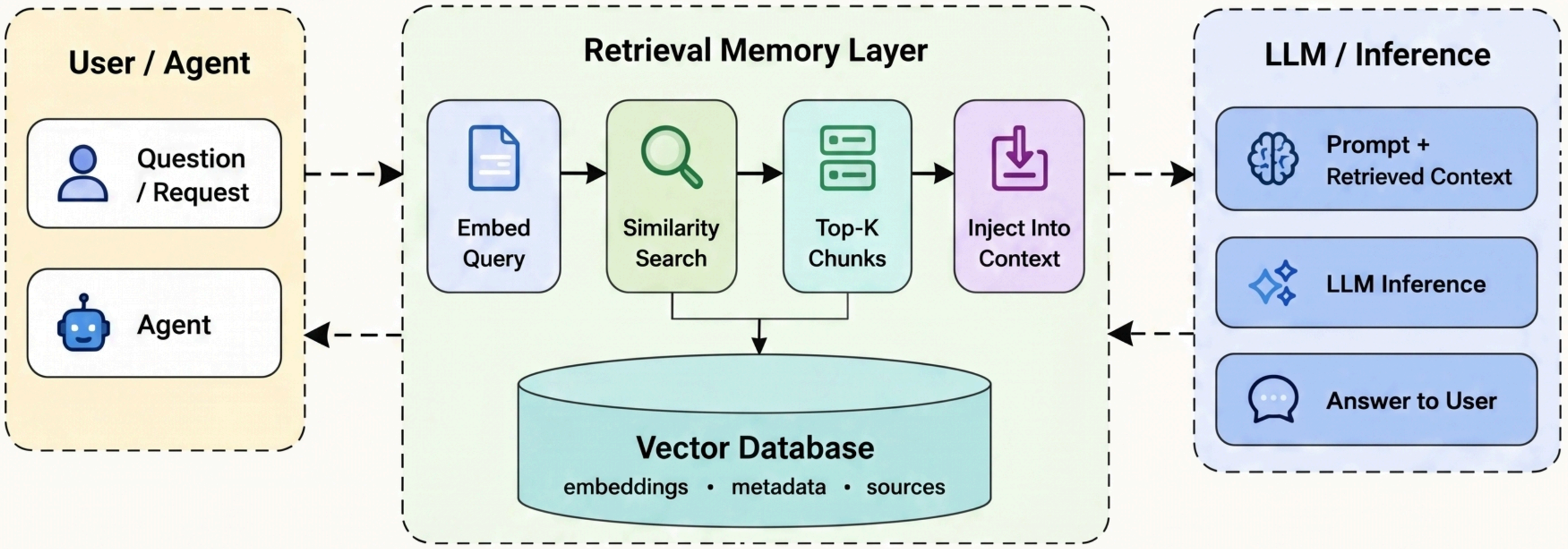
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"How do you design an agent that can learn and reuse procedures?"

→ Look for skill libraries, versioning, retrieval, and safe execution.

External / Retrieval Memory

Bring outside knowledge in.



WHAT IT IS Outside knowledge in a vector DB, fetched at inference via similarity search.

WHY IT'S USED Context window is finite; RAG gives access to vast, fresh information.

WHERE IT'S USED Knowledge-base Q&A, document analysis, support agents — lookup on demand.

EXAMPLE A support agent embeds your docs, stores them in a vector DB, and retrieves the most relevant chunks when a user asks.

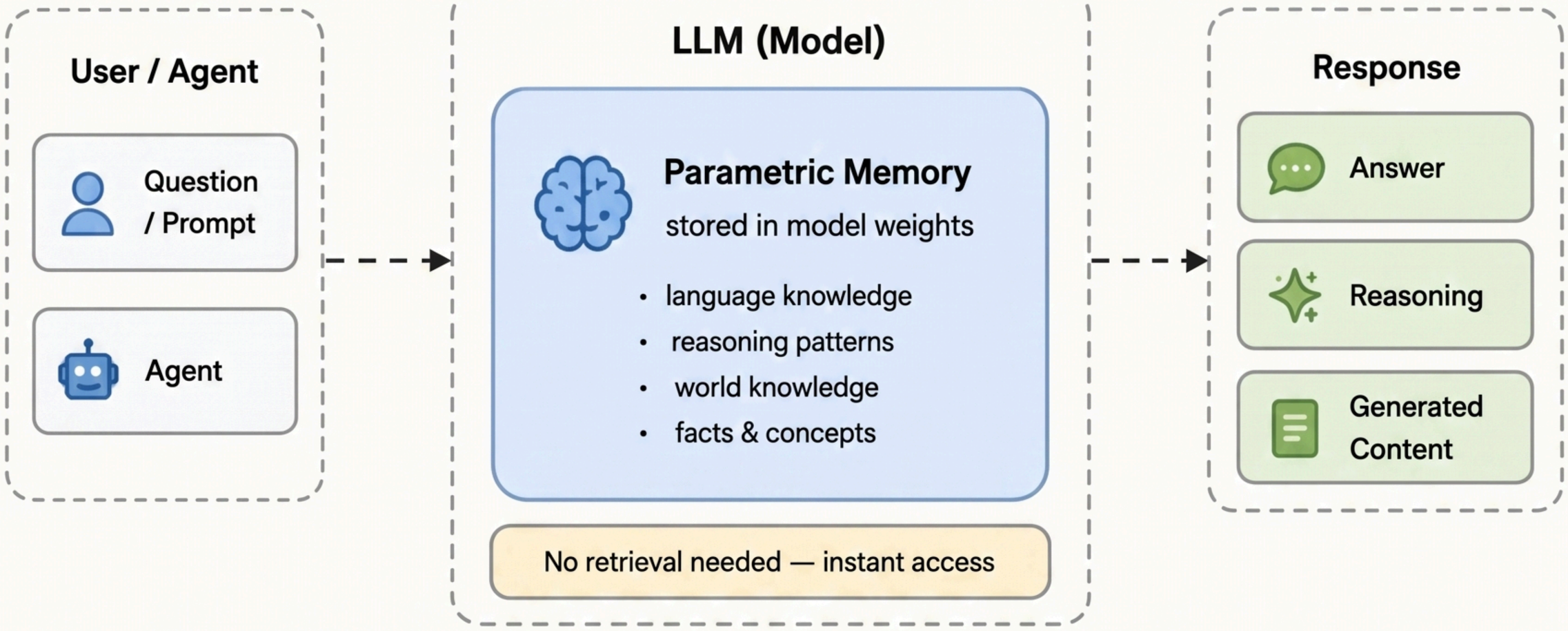
▶ INTERVIEW SIGNAL @DEVENDHIRAN

"How would you design retrieval for your support agent?"

→ Look for embedding strategy, chunking, indexing, similarity search, and freshness.

Parametric Memory

Knowledge baked into the weights.



WHAT IT IS Knowledge stored in model weights during training.

WHY IT'S USED Fastest and cheapest memory — no retrieval required.

WHERE IT'S USED Everywhere, implicilty. Limited by training cutoff; hard to update or audit.

EXAMPLE The model knows what a REST API is without being told.
But it won't know about a library released last month.

▸ **INTERVIEW SIGNAL**

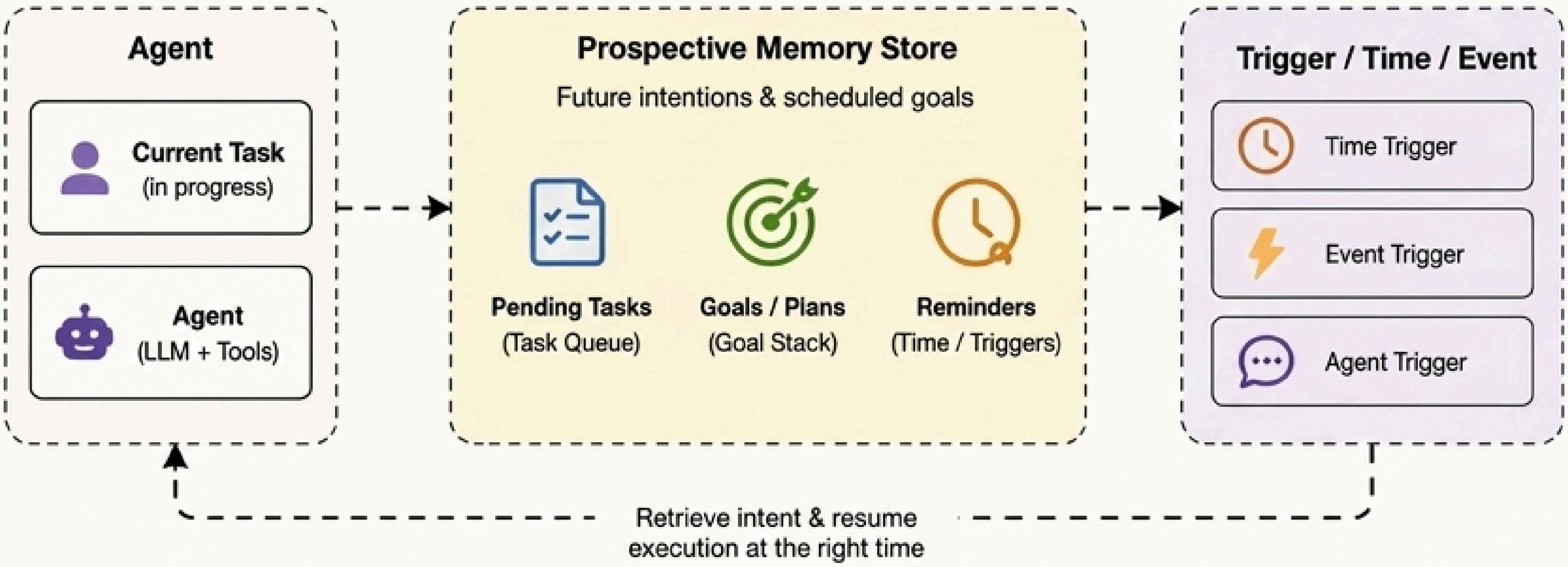
"How do you decide what to store in parametric memory vs. external memory?"

→ Great answers balance speed, cost, freshness, and verifiability.

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Prospective Memory

Remember what you plan to do next.



WHAT IT IS

A record of future intentions, scheduled goals, and follow-up actions the agent has committed to but not yet executed.

WHY IT'S USED

Offloads "what I need to do next" to a persistent store so long-horizon agents don't lose the thread when context or time shifts.

WHERE IT'S USED

Planning agents, autonomous task runners, multi-agent pipelines. Stored as task queues, goal stacks, or reminder logs alongside episodic memory.

EXAMPLE

Agent completes step 2 of 5, then gets interrupted.
It logs "resume at step 3 with these inputs."
When it restarts, it reads the intent and continues exactly where it left off.

▸ INTERVIEW SIGNAL

"How do you ensure your agent doesn't forget what it planned to do next?"

→ Use prospective memory: persist intents, set triggers, and resume execution at the right time.

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